LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Tuesday

SET 19: Survey Responses Analysis

Aim

To analyze survey responses using a grouped bar chart, stacked bar chart, table, and an interactive Tableau dashboard.

Dataset: Survey Responses

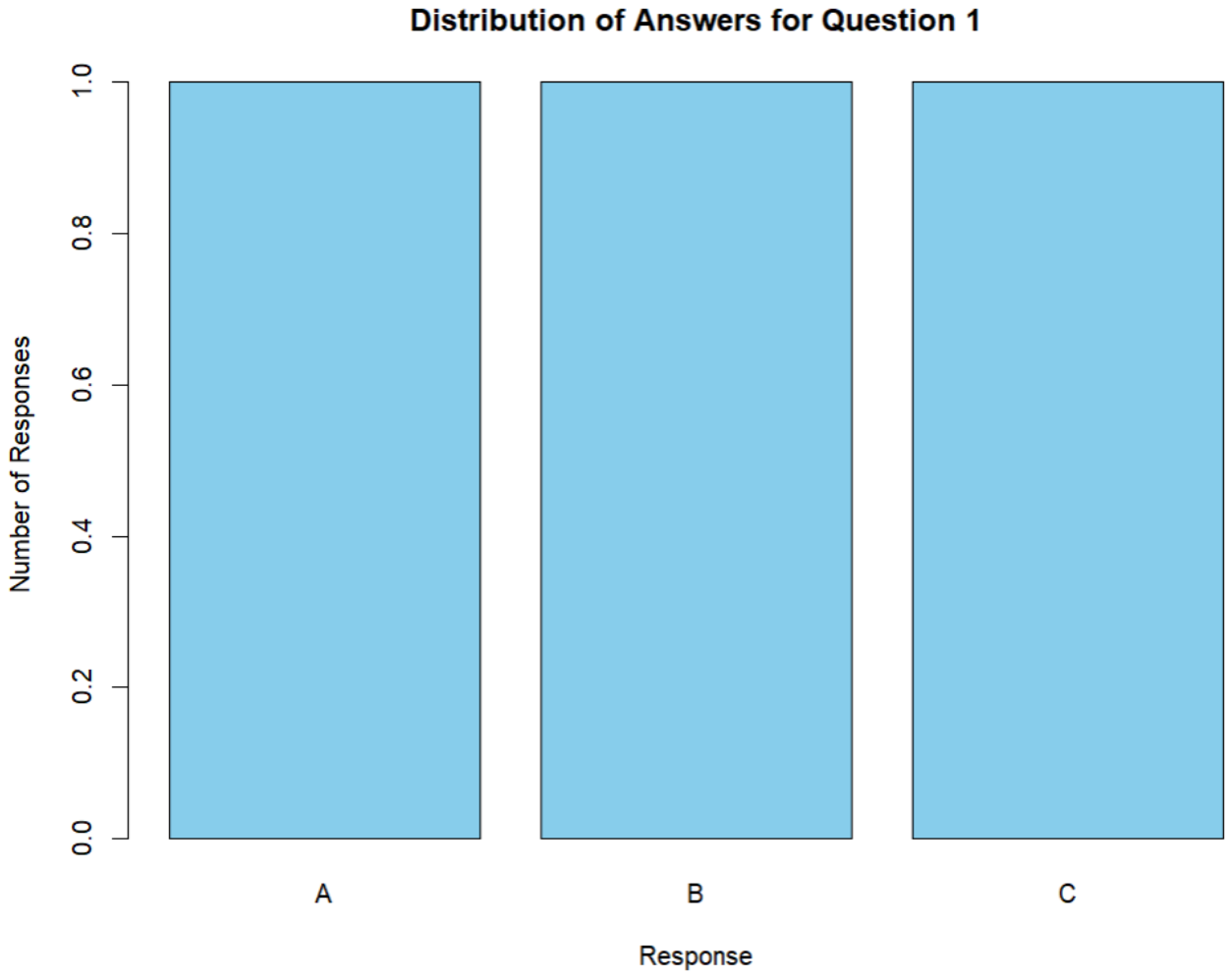
Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Q1. Create a grouped bar chart to visualize the distribution

R Program

```
q1 <- c("A", "B", "C")
q1_count <- table(q1)
barplot(
  q1_count,
  beside = TRUE,
  xlab = "Response",
  ylab = "Number of Responses",
  main = "Distribution of Answers for Question 1",
  col = "skyblue"
)
```

Output:



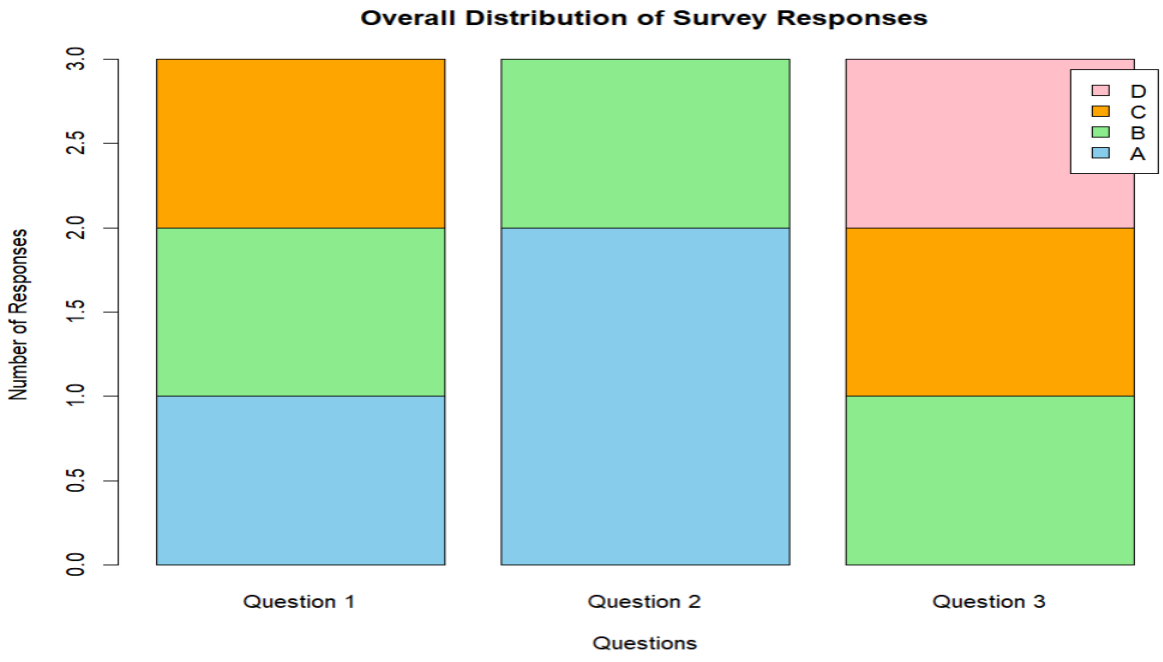
Observation: Each response occurs once in Question 1.

Q2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

R Program

```
q1 <- c("A", "B", "C")
q2 <- c("B", "A", "A")
q3 <- c("C", "D", "B")
responses <- rbind(
  "Question 1" = table(factor(q1, levels = c("A", "B", "C", "D"))),
  "Question 2" = table(factor(q2, levels = c("A", "B", "C", "D"))),
  "Question 3" = table(factor(q3, levels = c("A", "B", "C", "D")))
)
barplot(
  t(responses),
  beside = FALSE,
  names.arg = rownames(responses),
  xlab = "Questions",
  ylab = "Number of Responses",
  main = "Overall Distribution of Survey Responses",
  col = c("skyblue", "lightgreen", "orange", "pink"),
  legend.text = c("A", "B", "C", "D")
)
```

Output:



Observation: Question 1 has three different responses, Question 2 has two A responses and one B response, and Question 3 has one each of B, C, and D.

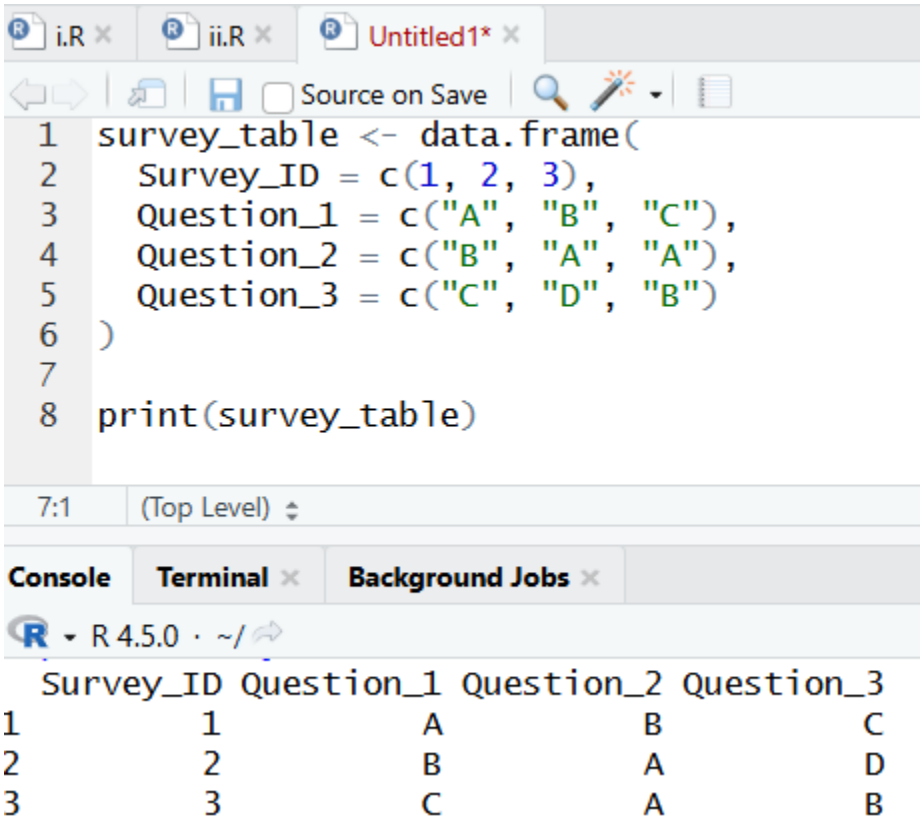
Q3. Build a table to show the survey response data for each survey.

R Program

```
survey_table <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

print(survey_table)
```

Output



Observation: The table displays the responses submitted by each survey participant.

Q4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and table.

Step 1 — Prepare the CSV dataset

Save the following content as Survey_Responses.csv.

Survey ID,Question 1,Question 2,Question 3
1,A,B,C
2,B,A,D
3,C,A,B

Step 2 — Import the dataset

1. Open Tableau.
2. Select Connect → Text File.
3. Import Survey_Responses.csv.
4. Open the Data Source page.

Step 3 — Create the grouped bar chart

1. Open a new worksheet and rename it Question 1 Distribution.
2. Drag Question 1 → Columns.
3. Drag Survey ID → Rows.
4. Click the dropdown on Survey ID and select Measure → Count.
5. Drag Question 1 → Color on the Marks card.
6. Choose Bar from the Marks dropdown.
7. Set the title to Distribution of Answers for Question 1.

Step 4 — Create the stacked bar chart

First, convert the three question columns into a long-format table.

1. Return to the Data Source page.
2. Select Question 1, Question 2, and Question 3.
3. Right-click and select Pivot.
4. Rename Pivot Field Names to Question.
5. Rename Pivot Field Values to Response.

Now create the chart:

1. Open a new worksheet named Overall Response Distribution.
2. Drag Question → Columns.
3. Drag Survey ID → Rows and change it to Count.
4. Drag Response → Color.
5. Select Analysis → Stack Marks → On.
6. Set the title to Overall Distribution of Survey Responses.

Step 5 — Create the response table

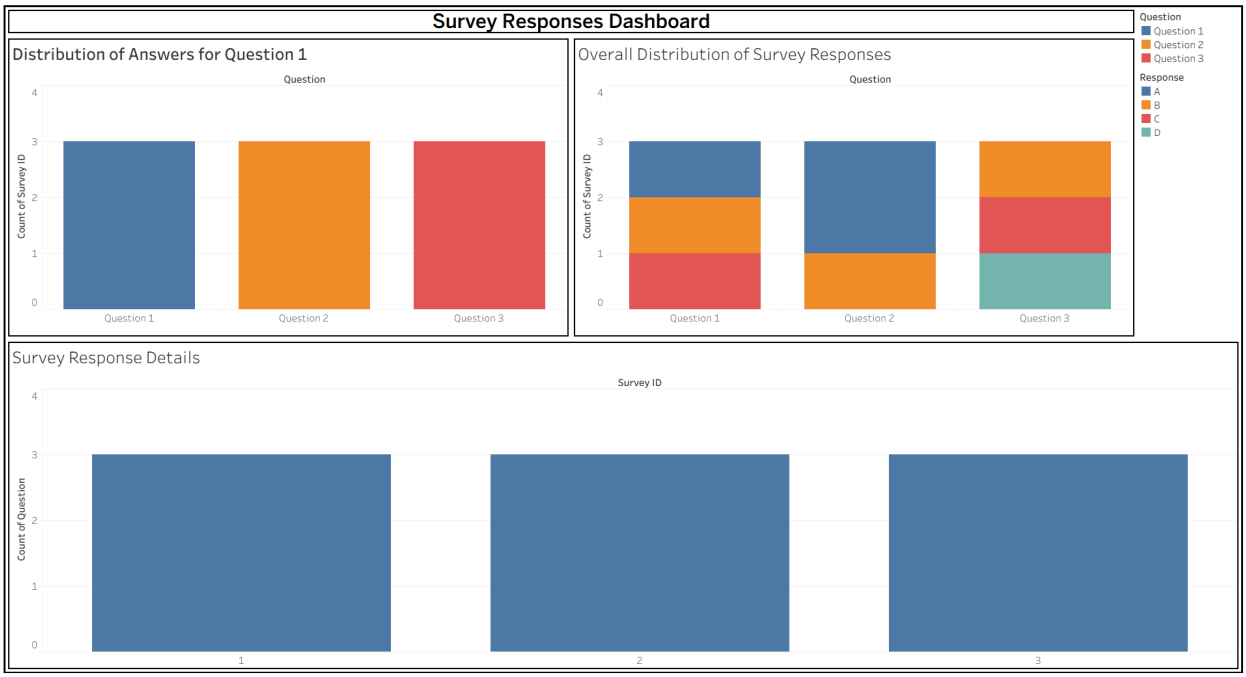
1. Create a new worksheet named Survey Response Table.
2. Drag Survey ID → Rows.
3. Add Question 1, Question 2, and Question 3 to the view.
4. Select Show Me → Text Table, if necessary.

5. Set the title to Survey Response Details.

Step 6 — Create the dashboard

1. Click New Dashboard.
2. Drag Question 1 Distribution to the top-left.
3. Drag Overall Response Distribution to the top-right.
4. Drag Survey Response Table to the bottom.
5. Add filters if required.
6. Set the dashboard title to Survey Responses Dashboard.

Expected dashboard



Procedure

1. Entered the survey response dataset.
2. Created a grouped bar chart, stacked bar chart, and response table using R.
3. Imported the dataset into Tableau.
4. Created individual worksheets for each visualization.
5. Combined the worksheets into a dashboard for interactive exploration.

Result

The survey responses were analyzed using bar charts and a table. A Tableau dashboard combining the Question 1 response distribution, overall response distribution, and survey response details was designed successfully.